LEARNER GUIDE

Surds: meaning and simplifying

Number Systems and Calculator Judgement · Lesson 5. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

√2 is the number which, multiplied by itself, gives exactly 2. Its decimal never ends or repeats, so it is **irrational**, and a root like this is called a **surd**. Rounding loses exactness: 1.414 × 1.414 = 1.999396, not 2. Surds can be tidied: 72 dots make 2 square blocks of 6 × 6, so √72 = √36 × √2 = 6√2. The **largest** square factor gives the simplest form in one step.

What you are learning to notice

- $\sqrt{n}$ is a whole number only when n is a square number. Otherwise it is a surd.
- $\sqrt{a} \times \sqrt{b} = \sqrt{(ab)}$ and $\sqrt{a} \div \sqrt{b} = \sqrt{(a \div b)}$. But $\sqrt{a} + \sqrt{b}$ is **not** $\sqrt{(a + b)}$: √9 + √16 = 7, while √25 = 5.
- To simplify, find the largest square factor: $\sqrt{(k^2 m)} = k\sqrt{m}$.
- Only like surds can be added: 2√3 + 5√3 = 7√3, but √2 + √3 stays as it is.
- Check a simplification by working back: 6√2 = √36 × √2 = √72.

Moving through it, one step at a time

List the square numbers: 4, 9, 16, 25, 36, 49, 64, 81, 100. Find the largest one that divides the number under the √. Take its root outside the √ and leave the other factor inside. Before adding or subtracting surds, simplify each one, then collect like surds.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — The square-factor finder

Simplify $\sqrt{72}$.

1. Try 4: $72 = 4 \times 18$, so $\sqrt{72} = 2\sqrt{18}$.	true, but $18 = 9 \times 2$ still has a square factor
2. Find the largest square factor of 72.	$72 = 36 \times 2$
3. 72 dots make 2 blocks of 6×6.	$\sqrt{72} = \sqrt{36} \times \sqrt{2}$
4. Take the root of 36 outside.	$\sqrt{72} = 6\sqrt{2}$
5. Check: $6^2 \times 2 = 36 \times 2 = 72$.	✓

$\sqrt{72} = 6\sqrt{2}$. The largest square factor, 36, does it in one step.

Example 2 — Adding surds

Work out $\sqrt{12} + \sqrt{75}$.

1. Simplify each surd first.	$\sqrt{12} = \sqrt{4} \times \sqrt{3} = 2\sqrt{3}$
2. Then the second one.	$\sqrt{75} = \sqrt{25} \times \sqrt{3} = 5\sqrt{3}$
3. Both are multiples of $\sqrt{3}$: like surds.	$2\sqrt{3} + 5\sqrt{3}$
4. Collect them, like $2x + 5x$.	$= 7\sqrt{3}$

$\sqrt{12} + \sqrt{75} = 7\sqrt{3}$. ($\sqrt{12} + \sqrt{75}$ is not $\sqrt{87}$.)

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (square numbers and factors), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.

- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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